Parallel Computing

MESTRADO EM ENGENHARIA INFORMÁTICA – ENGENHARIA DE DADOS
ANO LETIVO: 2025/2026

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- "If a single computer can solve a problem in N seconds, N computers can solve the same problem in 1 second?"

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PARALLELISM versus CONCURRENCY

- **Concurrency** is a property of a program (at design level) where two or more tasks can be in progress simultaneously.
- **Parallelism** is a run-time property where two or more tasks are being executed simultaneously

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PARALLEL COMPUTING

- Parallel and distributed programming is becoming the common programming paradigm, given the evolution of hardware to multicore architectures and massively parallel elements such as GPUs. Today's personal computer is made up of multiple processors that collectively provide greater processing power than previous single-core processors, but individually have less capacity. Programmers will need to master multiprocessor programming so that they can efficiently use the machines of today and tomorrow.
- Acquisition of knowledge leading to the simultaneous use of several processing units in a computer system. Building solid foundations in parallel architectures, parallelization of algorithms, programming models, process synchronization and performance measurements, through the development of programs.

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PARALELL COMPUTING

- Introduce the basic concepts and ideas in parallel computing and distributed Computing
- Advantages and limitations
- Parallel vs distributed computing
- Familiarize with the major programming models in parallel computing
- Provide with guidance for designing efficient parallel programs

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PARALLEL COMPUTING

The computation resources can include:

- A single computer with multiple processors
- An arbitrary number of computers connected by a network
- A combination of both

The computational problem usually demonstrates characteristics such as the ability to be:

- Decompose apart into discrete pieces of work that can be solved simultaneously
- Execute multiple program instructions at any moment in time
- Solved in less time with multiple compute resources than with a single computation resource

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WHY USE PARALLEL COMPUTING?

Real World Is Massively Complex

- In the real world, many complex, interrelated events are happening at the same time, within a temporal sequence.
- Compared to serial computing, parallel computing is much better suited for modeling, simulating and understanding complex, real world phenomena.

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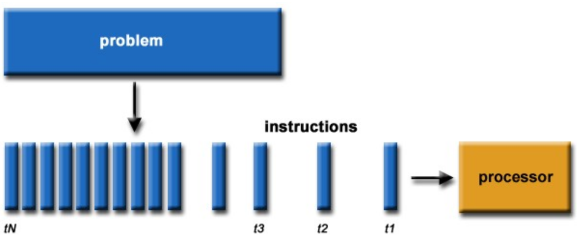
SERIAL COMPUTING

- Serial programming is the default method of code execution. Serial code is run sequentially; meaning, **only one instruction is processed at a time**.
- Traditionally, software has been written for ***serial*** computation:
  - A problem is broken into a discrete series of instructions
  - Instructions are executed sequentially one after another
  - Executed on a single computer having a single Central Processin[g] (CPU)
  - Only one instruction may execute at any moment in time T



https://hpc.llnl.gov/documentation/tutorials/introduction-parallel-computing-tutorial##Overview

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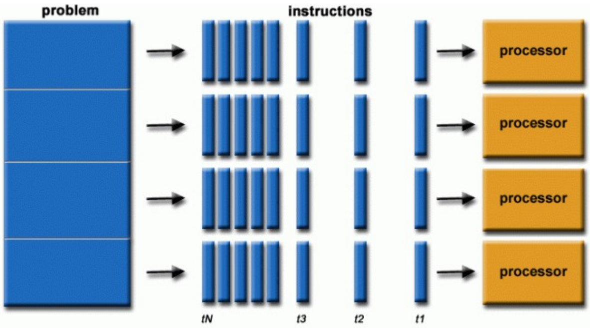
PARALLEL COMPUTING

Parallel computing is the simultaneous use of multiple compute resources to solve a computational problem:

- A problem is broken into discrete parts that can be solved concurrently
- Each part is further decomposed down to a series of instructions
- Instructions from each part execute simultaneously on multiple CPU
- An overall control/coordination mechanism is employed



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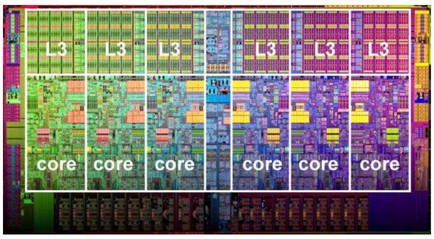
WHY USE PARALLEL COMPUTING?

- The following reasons to use parallel programming are:
  - Reduce the time it takes to troubleshoot a problem.
  - Solve more complex and larger problems.
  - Provide concurrency.

- Other reasons are:
  - Take advantage of computing resources that are not available locally or are underutilized.
  - Overcome memory limitations when memory is available on a single computer is insufficient to solve the problem.
  - To overcome the physical limits of speed and miniaturization that are currently beginning to restrict the possibility of building increasingly faster sequential computers.

Real World Is Massively Complex



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WHAT IS PARALLEL COMPUTING?

- **Parallel computing** lets you perform computational tasks using multiple processors simultaneously.
  - Tasks are divided into sub-tasks, which are then decomposed down further into instructions. Each instruction is then assigned to a different processor.
  - Processors connect via a shared memory space. The processors then complete these tasks simultaneously.
- **Parallel computing** is used for massive processing power and complex calculations. Parallel computing can help to carry out processes efficiently.

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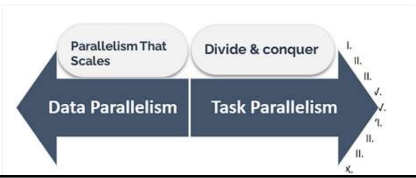
TYPES OF PARALLELISM:

- **Parallel computing** is a type of computation where multiple processes are executed simultaneously to solve a problem faster than traditional sequential computing. This is especially important as the size of data and the complexity of problems continue to grow, necessitating more efficient processing techniques.
- Parallelism in computing can be understood as the ability to perform multiple operations concurrently.
  - The main aim is to increase computational speed and efficiency.
  - By dividing tasks into smaller subtasks and executing them at the same time, computers can solve complex problems more quickly than by executing them one after the other.
- Parallelism can be classified broadly into two categories: **data parallelism** and **task parallelism**. These two types differ in how they utilize processing resources and manage data.



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TYPES OF PARALLELISM: **DATA PARALLELISM**

Data parallelism focuses on distributing data across multiple processors. Each processor performs the same operation on different pieces of data simultaneously. This approach is effective in scenarios where large datasets are involved and the same computation needs to be applied repeatedly to different data elements.

- **Single Instruction, Multiple Data (SIMD)**: Data parallelism typically follows the SIMD model, where a single operation is applied to multiple data points simultaneously. This is common in graphics processing and scientific computations.
- **Homogeneous Operations**: The operations performed on the data are the same, which simplifies the design of parallel algorithms.
- **Scalability**: Data parallelism is highly scalable, meaning that as more processors are added, the computation can be efficiently divided among them, improving performance.

Examples of Data Parallelism:

- **Vector Operations**
- **Image Processing**
- **Machine Learning**

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TYPES OF PARALLELISM: **DATA PARALLELISM**

Task parallelism involves distributing different tasks or operations across multiple processors. Unlike data parallelism, where the same operation is performed on different data elements, task parallelism allows for different operations to be executed at the same time. This is useful when tasks are independent or when they need to be executed in a certain order.

- Multiple Instructions, Multiple Data (MIMD): Task parallelism typically follows the MIMD model, where different processors execute different instructions on different data.
- Heterogeneous Operations: The tasks can be entirely different operations, allowing for more complex workflows.
- Dynamic Work Distribution: Tasks can be dynamically assigned to processors based on availability and workload, which can lead to better resource utilization.

Examples of Task Parallelism:

- **Web Servers**
- **Simulations**
- **Build Systems**

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PARALLEL COMPUTING ADVANTAGES

- **Speed** - Parallel computing is able to perform computations much faster than traditional, serial computing. This is because it processes multiple instructions simultaneously using different processors. The more processors available, the faster the speed. Problem-solving faster by reducing the time taken to get results. It also increases the speed of decision-making.
- **Higher Throughput** - Parallel computing executes tasks and processes concurrently. This increases the throughput of the system. This is important in scenarios where a high volume of data needs processing in a limited amount of time.
- **Scalability** - A parallel computing system can be easily scaled by adding or removing processors. The overall computational power of a system can increase or decrease as necessary. If more computational power is suddenly needed, the system can expand as required. Think of it as akin to a call center, where new phone lines can be added to a PBX system to handle higher volumes of calls.
- **Better Resource Utilization** - Parallel computing systems distribute their workload across the hardware resources available to them. This limits the overutilization or underutilization of specific resources. This improves the overall efficiency of the system.

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PARALLEL COMPUTING DISADVANTAGES

- **Initial Cost** - Parallel computing may require specialized software and hardware to fully realize its potential. This can result in larger initial costs when first setting up the system. Costs can increase if additional processors are needed to scale the system.
- **Complexity** - The algorithms needed for parallel computing are more complex than serial computing. This makes it harder to manage data distribution and communication between parallel processors. It's also more difficult to debug parallel computing solutions than their serial alternatives.
- **Performance** - Parallel computing often requires synchronization and communication mechanisms between processors to ensure consistency. Using these mechanisms can raise overheads, and create issues with network latency. This can work to reduce the performance gains in some systems.

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WHEN TO USE PARALLEL COMPUTING ?

Parallel computing is commonly used for complex computational problems, which can be decomposed into smaller/simpler tasks. Executing these smaller tasks simultaneously leads to:

- Faster execution times
- Improved performance
- Better resource utilization

- Scenarios which often involve the use of parallel computing, include:
  - Large-scale data processing, e.g. big data analytics
  - Scientific simulations and modeling, e.g. climate modeling, bioinformatics
  - Image processing, e.g. medical imaging
  - Video processing, e.g. video color correction
  - Machine learning, e.g. financial risk management algorithms
  - Video game development, e.g. computer physics simulations

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DISTRIBUTED COMPUTING

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WHAT IS DISTRIBUTED COMPUTING?

- **Distributed computing** connects computers so that they can act as one powerful machine.
- They work together to perform complex computations. This allows them to perform tasks that no single computer in the network would be able to achieve.
- Computers are either connected through a local network if they are geographically close to one another, or through a wide area network (WAN) if they are geographically distant.
- A distributed system is made up of a variety of different devices, such as computers, mainframes, and minicomputers. Each device within the system is referred to as a 'node', with a group of nodes known as a 'cluster.'
- Nodes in the system communicate with one another by passing messages through the network. It's even possible to remotely access and control other nodes within a cluster using a remote desktop management tool.
- During distributed computing, the various steps in a process will be distributed to the most efficient machine in the network.
  - The user interface processing will occur on the computer being accessed by the user, while application processing will occur on a remote machine.
  - Database accessing and processing algorithms will take place on another remote machine. This is usually one that can provide centralized access for a range of processes.

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ADVANTAGES OF DISTRIBUTED COMPUTING

- **Flexibility & Adaptability** - Distributed systems can change to meet changing requirements. It's possible to add or remove new nodes as necessary. Developers can change or reconfigure the system entirely as workloads change. Distributed computing systems are well-suited to organizations that have varying workloads.
- **Wide Distribution** - Using distributed computing, organizations can create systems that span multiple geographic locations. This is perfect for multinational corporation systems. These often require collaboration between users in different locations, using any number of domains from Only Domains. Distributed computing facilitates global collaboration. Users in different geographic locations can access and contribute to shared resources.
- **Data Redundancy & Backup** - A distributed system will often replicate data across multiple nodes. This helps to provide data redundancy and backup capabilities. In the event of a disruption such as hardware failure, data availability will still be maintained thanks to the other nodes in the system.
- **Performance** - Like parallel computing, distributed computing enables the parallel execution of tasks. A distributed system provides improved performance and faster execution times. It does this by carrying out tasks across multiple nodes.

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DISADVANTAGES OF DISTRIBUTED COMPUTING

- **Security Concerns** - Distributed computing systems provide a greater number of access points for malicious actors. Any weak point presents a security threat to the entire system. Robust security measures must be put in place to ensure secure communication and that data is protected across the entire system.
- **Software Complexity & Compatibility** - Writing code that will function correctly in a distributed computing environment can be tricky, as it must be able to handle issues such as data partitioning, synchronization, and task distribution. It can also be challenging to ensure that the software running on different nodes is all compatible with one another. Compatibility issues can arise from differences in operating systems, libraries, and hardware configurations.
- **Resource Management** - Resource allocation and load balancing can present a challenge for distributed computing systems. Nodes may feature varying levels of computing power, storage, and memory. This can lead to issues with performance, such as increased latency. This impacts the responsiveness of distributed systems. This is especially the case with tasks that require frequent internode communication.

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WHEN TO USE DISTRIBUTED COMPUTING?

- **Distributed computing** is often used to build and deploy powerful applications. It can also be used when a computational task can be divided into smaller components. It's especially useful in scenarios that inlude:
  - High scalability
  - Frequent resource sharing
  - High fault tolerance
  - Extensive collaboration
- Scenarios that often involve the use of distributed computing include:
  - Communication networks, e.g. a VoIP phone system
  - Data processing, e.g. analyzing data from multiple data repositories
  - High traffic web services and applications, e.g. search engines
  - Content streaming services, e.g. Netflix, Disney +
  - Internet of Things (IoT) devices

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PARALLEL VS. DISTRIBUTED COMPUTING

Parallel computing	Distributed computing
Different types of processes can be done simultaneously.	The location of system components is different and a single task is distributed among the systems.
A single computer can be used for parallel computing.	A distributed computing system needs a lot of computers.
Parallel system consists of multiple processors which help in the performance of multiple operations.	Multiple operations are performed by multiple computers in distributed computing.
Parallel computing supports shared or distributed memory.	Distributed computing has distributed memory.
The bus is used for communication between different processors.	Message passing is used for communication between computers.

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PARALLEL VS. DISTRIBUTED COMPUTING

Parallel computing	Distributed computing
System performance can be improved easily through parallel computing.	•Distributed computing is able to perform the following: •System scalability improvement •Fault tolerance •Resource sharing capabilities
A single thread can be used to manage all the tasks.	Computers in a distributed system coordinate with each other by using advanced mechanisms.
Scalability is limited as the number of processors in a system is limited.	High scalability is offered as more computers can be added to a network if needed.
Fault tolerance is limited.	Fault tolerance is comparatively much more as handling network issues and node failures is easy.
A single master clock is shared by all the processors.	This system uses synchronization algorithms.

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PARALLEL VS. DISTRIBUTED COMPUTING

Parallel computing	Distributed computing
Many operations are performed simultaneously	System components are located in different locations
A single computer is required	Uses multiple computers
Multiple processors perform various operations	Multiple computers perform various operations
Can have shared or distributed memory	It only has distributed memory
Processors communicate with each other via bus	Computers communicate with each other through message exchange.
Improves system performance	Improves system scalability, fault tolerance, and resource-sharing capabilities

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